

Candidate Assessment

Role: Data Scientist – IFTA Evidence Normalization, Analytics & Machine Learning

This assessment is designed with an AWS-based IFTA audit environment in mind. Candidates are therefore encouraged to demonstrate their solution using relevant AWS services (e.g., Amazon S3, Textract, Glue, Athena, SageMaker, Lambda, Step function, MLFlow).

About IFTA

The International Fuel Tax Agreement (IFTA) is an agreement among jurisdictions in Canada and the United States that simplifies fuel tax reporting for motor carriers operating across multiple jurisdictions. IFTA compliance is based on reconciling fuel purchase records with distance traveled by jurisdiction. Analytics and machine learning can help identify anomalous or high-risk records for audit and compliance review by detecting inconsistencies, implausible usage patterns, and gaps that are difficult to identify through manual review alone.

Context

You are provided with fuel invoice data and distance logs, representing IFTA operational evidence. The data contains noise, inconsistencies, and gaps, which are intentional. You may assume the raw data is already accessible in AWS (e.g., S3), but not fully ML-ready.

Objective

Prepare ML-ready datasets and apply analytical or machine learning techniques to:

- identify anomalous fuel or distance patterns
- support audit risk detection
- explain findings clearly and defensibly

Scope of Work

1. Data Curation & Preparation

- Extract, clean and standardize dates, locations, and units
- Join fuel and distance data where appropriate
- Handle missing and inconsistent values transparently
- Document your assumptions and limitations

Note:

Extract the following data elements from invoices and distance logs where available:

- invoice / receipt number, transaction date, transaction time, vendor name, location (city, province), fuel type, litres purchased, price per litre, total cost, tax breakdown, form of payment, trip date, trip origin, trip destination, trip start and end time, start and end odometer, distance (km), VIN or truck number, distance traveled by jurisdiction.

- For each extracted record, retain a reference to the original source file and compute an extraction confidence score to assess data reliability. Standard OCR techniques may be used; however, AI-enabled OCR technologies, such as Amazon Textract, can further support automated data extraction, confidence scoring, and end-to-end audit traceability.

2. Feature Engineering

Engineer features such as:

- fuel litres per km
- fuel cost per km
- trips without nearby fuel purchases
- jurisdiction-level distance vs fuel proportions
- temporal trends (daily, weekly, monthly)

Explain why each feature is relevant in an IFTA context.

3. ML or Analytical Task

Apply one or more approaches:

- statistical outlier detection
- rule-based + ML anomaly detection
- clustering of fuel/distance behavior

Please note, model sophistication is less important than reasoning and clarity.

4. Interpretation & Explainability

- Clearly explain why records are flagged
- Discuss model limitations and data quality impacts
- Relate findings to audit relevance

5. AWS Usage

- Use AI to solve the puzzle, explore AWS such as S3, Textract, SageMaker, Athena / Glue or notebooks or anything you like and justify for data extraction, curation, analysis and ML (deployment not required)

Deliverables

- Analysis notebooks or scripts
- Description of ML-ready dataset
- Feature list with rationale
- Analytical / ML results
- Findings, interpretation, assumptions, and limitations

Interview Presentation (maximum 20 minutes)

You will walk the interviewers through:

- problem understanding
- data extraction and preparation pipeline (quality issues, null handling and suggestions to improve)

- approach and workflow for ML ready dataset
- ML modelling
- effective use of AWS service in the workflow [justification, scalability and maintainability etc]
- a live demonstration of what you built and which AWS services you used.

Evaluation Rubric (100 points total)

Category	Points	What We Look For
Problem understanding	15	Clarity, purpose & audit needs
Data preparation	25	Use of technology, confidence, gaps & null handling
Feature engineering	15	Clarity, importance and relevance to the problem
Code & reproducibility	15	Organization, clarity and ability to re-run
Communication & presentation	30	Narrative, suggestion and decision-making support